

$$\begin{aligned}
 125. \text{ Given exp. } &= \frac{(1+\sqrt{3})(1-\sqrt{3})-(1-\sqrt{3})+(1+\sqrt{3})}{(1+\sqrt{3})(1-\sqrt{3})} \\
 &= \frac{1-(\sqrt{3})^2-1+\sqrt{3}+1+\sqrt{3}}{1^2-(\sqrt{3})^2} \\
 &= \frac{2\sqrt{3}-2}{1-3} = \frac{2(\sqrt{3}-1)}{(-2)} = 1-\sqrt{3}.
 \end{aligned}$$

$$\begin{aligned}
 126. \text{ Given exp. } &= (2+\sqrt{2}) + \frac{1}{(2+\sqrt{2})} \times \frac{(2-\sqrt{2})}{(2-\sqrt{2})} \\
 &\quad - \frac{1}{(2-\sqrt{2})} \times \frac{(2+\sqrt{2})}{(2+\sqrt{2})} \\
 &= (2+\sqrt{2}) + \frac{(2-\sqrt{2})}{(4-2)} - \frac{(2+\sqrt{2})}{(4-2)} \\
 &= (2+\sqrt{2}) + \frac{1}{2}(2-\sqrt{2}) - \frac{1}{2}(2+\sqrt{2}) = 2.
 \end{aligned}$$

$$\begin{aligned}
 127. \text{ Given exp. } &= \frac{1}{(\sqrt{5}+\sqrt{3})} \times \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})} + \frac{2}{(3+\sqrt{5})} \\
 &\quad \times \frac{(3-\sqrt{5})}{(3-\sqrt{5})} - \frac{3}{(3+\sqrt{3})} \times \frac{(3-\sqrt{3})}{(3-\sqrt{3})} \\
 &= \frac{(\sqrt{5}-\sqrt{3})}{(5-3)} + \frac{2(3-\sqrt{5})}{(9-5)} - \frac{3(3-\sqrt{3})}{(9-3)} \\
 &= \frac{\sqrt{5}-\sqrt{3}}{2} + \frac{2(3-\sqrt{5})}{4} - \frac{3(3-\sqrt{3})}{6} \\
 &= \frac{6(\sqrt{5}-\sqrt{3})+6(3-\sqrt{5})-6(3-\sqrt{3})}{12} = 0.
 \end{aligned}$$

$$\begin{aligned}
 128. \frac{7}{(3+\sqrt{2})} &= \frac{7}{(3+\sqrt{2})} \times \frac{(3-\sqrt{2})}{(3-\sqrt{2})} = \frac{7(3-\sqrt{2})}{(9-2)} \\
 &= (3-\sqrt{2}) = (3-1.4142) = 1.5858.
 \end{aligned}$$

$$\begin{aligned}
 129. \text{ Given exp. } &= \frac{3\sqrt{2}}{(\sqrt{6}-\sqrt{3})} \times \frac{(\sqrt{6}+\sqrt{3})}{(\sqrt{6}+\sqrt{3})} - \frac{4\sqrt{3}}{(\sqrt{6}-\sqrt{2})} \\
 &\quad \times \frac{(\sqrt{6}+\sqrt{2})}{(\sqrt{6}+\sqrt{2})} - \frac{6}{2(\sqrt{2}-\sqrt{3})} \\
 &= \frac{3\sqrt{2}(\sqrt{6}+\sqrt{3})}{(6-3)} - \frac{4\sqrt{3}(\sqrt{6}+\sqrt{2})}{(6-2)} \\
 &\quad + \frac{3}{(\sqrt{3}-\sqrt{2})} \times \frac{(\sqrt{3}+\sqrt{2})}{(\sqrt{3}+\sqrt{2})} \\
 &= \sqrt{2}(\sqrt{6}+\sqrt{3}) - \sqrt{3}(\sqrt{6}+\sqrt{2}) + 3(\sqrt{3}+\sqrt{2}) \\
 &= \sqrt{12} + \sqrt{6} - \sqrt{18} - \sqrt{6} + 3\sqrt{3} + 3\sqrt{2} \\
 &= 2\sqrt{3} - 3\sqrt{2} + 3\sqrt{3} + 3\sqrt{2} = 5\sqrt{3}
 \end{aligned}$$

$$\begin{aligned}
 130. \frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} &= \frac{(\sqrt{7}+\sqrt{5})}{(\sqrt{7}-\sqrt{5})} \times \frac{(\sqrt{7}+\sqrt{5})}{(\sqrt{7}+\sqrt{5})} = \frac{(\sqrt{7}+\sqrt{5})^2}{(7-5)} \\
 &= \frac{7+5+2\sqrt{35}}{2} = \frac{12+2\sqrt{35}}{2} = 6+\sqrt{35}.
 \end{aligned}$$

$$131. a+b\sqrt{3} = \frac{(5+2\sqrt{3})}{(7+4\sqrt{3})} \times \frac{(7-4\sqrt{3})}{(7-4\sqrt{3})} = \frac{35-20\sqrt{3}+14\sqrt{3}-24}{(7)^2-(4\sqrt{3})^2}$$

$$= \frac{11-6\sqrt{3}}{49-48} = 11-6\sqrt{3}. \quad \therefore a = 11, b = -6.$$

$$\begin{aligned}
 132. \frac{\sqrt{2}-1}{\sqrt{2}+1} &= \frac{(\sqrt{2}-1)}{(\sqrt{2}+1)} \times \frac{(\sqrt{2}-1)}{(\sqrt{2}-1)} = (\sqrt{2}-1)^2. \\
 \therefore \sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1}} &= (\sqrt{2}-1) = (1.414-1) = 0.414.
 \end{aligned}$$

$$\begin{aligned}
 133. \text{ Given exp. } &= \frac{3+\sqrt{6}}{5\sqrt{3}-4\sqrt{3}-4\sqrt{2}+5\sqrt{2}} = \frac{(3+\sqrt{6})}{(\sqrt{3}+\sqrt{2})} \\
 &= \frac{(3+\sqrt{6})}{(\sqrt{3}+\sqrt{2})} \times \frac{(\sqrt{3}-\sqrt{2})}{(\sqrt{3}-\sqrt{2})} \\
 &= \frac{3\sqrt{3}-3\sqrt{2}+3\sqrt{2}-2\sqrt{3}}{(3-2)} = \sqrt{3} = 1.732.
 \end{aligned}$$

$$\begin{aligned}
 134. \text{ Given exp. } &= \frac{(2+\sqrt{3})}{(2-\sqrt{3})} \times \frac{(2+\sqrt{3})}{(2+\sqrt{3})} + \frac{(2-\sqrt{3})}{(2+\sqrt{3})} \times \frac{(2-\sqrt{3})}{(2-\sqrt{3})} \\
 &\quad + \frac{(\sqrt{3}-1)}{(\sqrt{3}+1)} \times \frac{(\sqrt{3}-1)}{(\sqrt{3}-1)} \\
 &= \frac{(2+\sqrt{3})^2}{(4-3)} + \frac{(2-\sqrt{3})^2}{(4-3)} + \frac{(\sqrt{3}-1)^2}{(3-1)} \\
 &= [(2+\sqrt{3})^2 + (2-\sqrt{3})^2] + \frac{4-2\sqrt{3}}{2} \\
 &= 2(4+3) + 2 - \sqrt{3} = 16 - \sqrt{3}.
 \end{aligned}$$

$$\begin{aligned}
 135. x+y &= \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} + \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}} = \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}-\sqrt{3})} \times \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}+\sqrt{3})} \\
 &\quad + \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}+\sqrt{3})} \times \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})} \\
 &= \frac{(\sqrt{5}+\sqrt{3})^2}{(\sqrt{5})^2-(\sqrt{3})^2} + \frac{(\sqrt{5}-\sqrt{3})^2}{(\sqrt{5})^2-(\sqrt{3})^2} \\
 &= \frac{(\sqrt{5}+\sqrt{3})^2 + (\sqrt{5}-\sqrt{3})^2}{5-3} \\
 &= \frac{2[(\sqrt{5})^2 + (\sqrt{3})^2]}{2} = 5+3 = 8.
 \end{aligned}$$

$$\begin{aligned}
 136. \text{ Given exp. } &= \frac{(\sqrt{2})^2}{(\sqrt{3})^2} \cdot \frac{(2+\sqrt{3})(2-\sqrt{3})}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{2}{3} \left[ \frac{2^2-(\sqrt{3})^2}{(\sqrt{3})^2-1^2} \right] \\
 &= \frac{2}{3} \left( \frac{4-3}{3-1} \right) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}.
 \end{aligned}$$

$$\begin{aligned}
 137. \text{ Given exp. } &= \frac{12}{3+(\sqrt{5}+2\sqrt{2})} \times \frac{3-(\sqrt{5}+2\sqrt{2})}{3-(\sqrt{5}+2\sqrt{2})} \\
 &= \frac{12(3-\sqrt{5}-2\sqrt{2})}{3^2-(\sqrt{5}+2\sqrt{2})^2} = \frac{12(3-\sqrt{5}-2\sqrt{2})}{9-(5+8+4\sqrt{10})} \\
 &= \frac{12(3-\sqrt{5}-2\sqrt{2})}{(-4-4\sqrt{10})} = \frac{3(\sqrt{5}+2\sqrt{2}-3)}{\sqrt{10}+1} \\
 &= \frac{3(\sqrt{5}+2\sqrt{2}-3)}{\sqrt{10}+1} \times \frac{\sqrt{10}-1}{\sqrt{10}-1}
 \end{aligned}$$